

General Intelligence Research Report

ATLAS Synthesis Report • Generated: Sep 19, 2026, 9:53 PM

General Intelligence Research Report

Date: September 19, 2026 (UTC)

Executive Summary

General intelligence, often denoted as *g*, is a psychometric construct that captures the common variance across diverse cognitive tasks. It accounts for roughly 40-50% of individual differences in mental performance and is widely used in intelligence testing, educational assessment, and occupational selection. This report synthesizes foundational theory, measurement approaches, major findings, contemporary debates, and emerging research directions up to 2026.

1. Foundations of General Intelligence

- Spearman's Two-Factor Theory (1904): Introduced the idea of a single general factor (*g*) underlying all mental abilities, plus specific factors (*s*) for individual tasks.
- *g* Factor (Psychometrics): Statistical construct derived from positive correlations among cognitive test scores. Frequently equated with IQ, general cognitive ability, or mental ability.
- Key Researchers: Charles Spearman, Raymond Cattell (fluid vs. crystallized intelligence), John B. Carroll (Three-Stratum Theory), Arthur Jensen, Richard Herrnstein, and more recent contributors like Ian Deary and James Flynn.

2. Measurement and Assessment

- IQ Tests: Stanford-Binet, Wechsler Adult Intelligence Scale (WAIS), Woodcock-Johnson Tests of Cognitive Abilities.
- Factor Analytic Methods: Principal component analysis, exploratory and confirmatory factor analysis to extract *g*.
- Alternative Metrics: Raven's Progressive Matrices (fluid intelligence), Cattell-Horn-Carroll (CHC) model, and modern computer-adaptive testing.

3. Empirical Findings

- Heritability: Twin and genome-wide association studies (GWAS) estimate heritability of *g* at 50-80% in adulthood.
- Environmental Influences: Early childhood nutrition, education quality, socioeconomic status, and enriched environments affect *g* development.
- The Flynn Effect: Global rise in IQ scores throughout the 20th century; recent data suggest a plateau or slight decline in some regions after 2010.
- Neurobiological Correlates: Structural MRI shows *g* correlates with total brain volume, prefrontal cortex thickness, and white-matter integrity. Functional connectivity patterns in the frontoparietal network also predict *g*.

4. Contemporary Debates

| Debate | Position A | Position B |

|---|---|---|

| Nature vs. Nurture | *g* is primarily genetic. | Environment can substantially modify *g* across the lifespan. |

| Single vs. Multiple Intelligences | *g* captures the core of intelligence. | Howard Gardner's theory of multiple intelligences argues for distinct, domain-specific abilities not reducible to *g*. |

| Cultural Fairness | Standardized IQ tests are culturally biased. | Modern adaptive tests mitigate bias; g remains a valid cross-cultural construct. |

| Predictive Validity | g predicts academic and occupational outcomes robustly. | Some argue that non-cognitive traits (personality, motivation) are equally important. |

5. Emerging Research (2020-2026)

1. Polygenic Scores: Large-scale GWAS (e.g., UK Biobank, 2022) produce polygenic indices explaining up to 12% of variance in g.
2. Neuroimaging-Genetics Integration: Studies linking specific SNPs to brain network efficiency associated with g.
3. Lifelong Cognitive Plasticity: Intervention trials (e.g., working-memory training, bilingualism) show modest but reliable gains in fluid intelligence.
4. AI-Assisted Assessment: Machine-learning models generate adaptive test items in real-time, improving measurement precision.
5. Cross-Cultural Mega-Studies: The International Cognitive Ability Resource (ICAR) project aggregates data from >100,000 participants across 50+ countries.

6. Practical Implications

- Education: Early identification of g disparities can guide targeted interventions.
- Employment: g remains a strong predictor of job performance; however, holistic assessment models incorporate personality and social skills.
- Public Policy: Understanding the determinants of g informs policies on early childhood education, nutrition, and socioeconomic inequality.

7. Future Directions

- Integrative Models: Combining genetic, neurobiological, and environmental data into unified predictive frameworks.
- Dynamic Assessment: Real-time analytics to track cognitive development longitudinally.
- Ethical Considerations: Managing privacy and potential misuse of polygenic scores and AI-driven assessments.

Prepared by an AI research assistant using up-to-date sources as of September 2026.